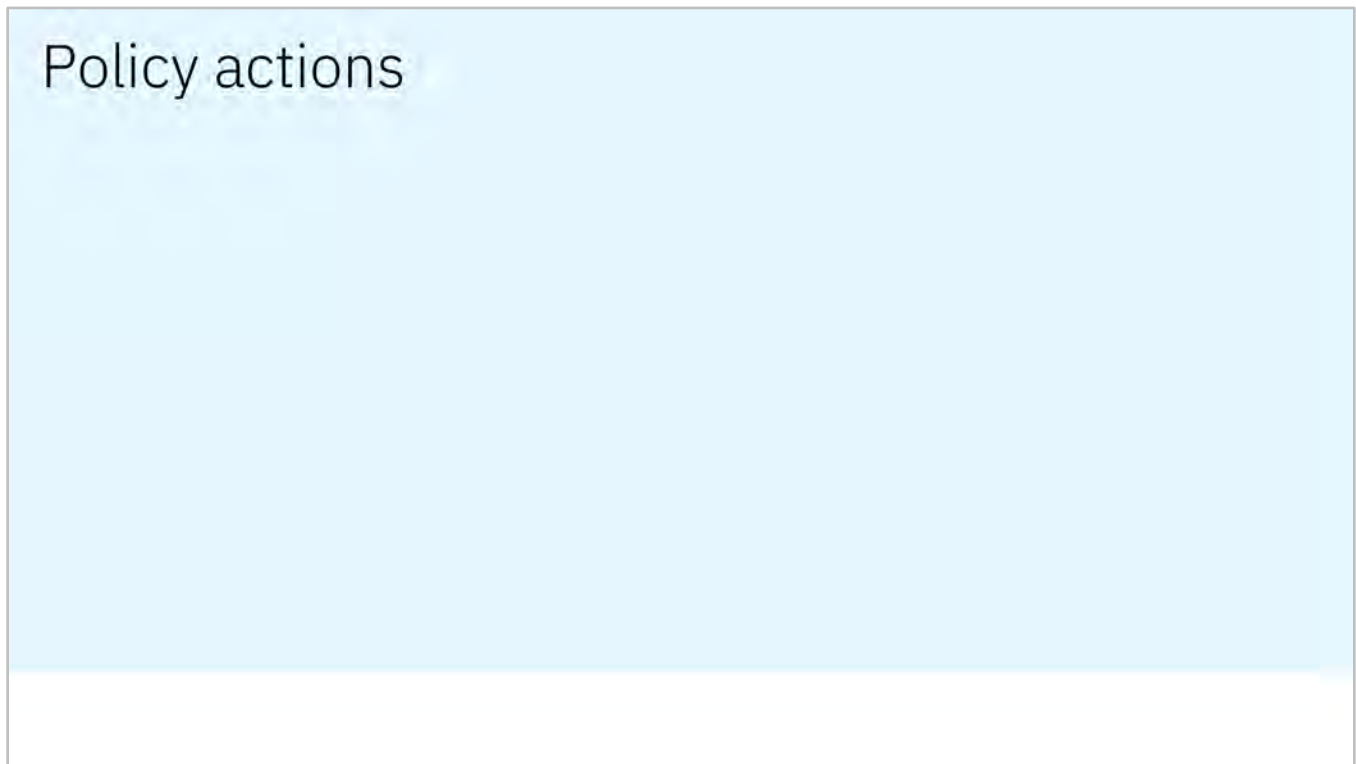


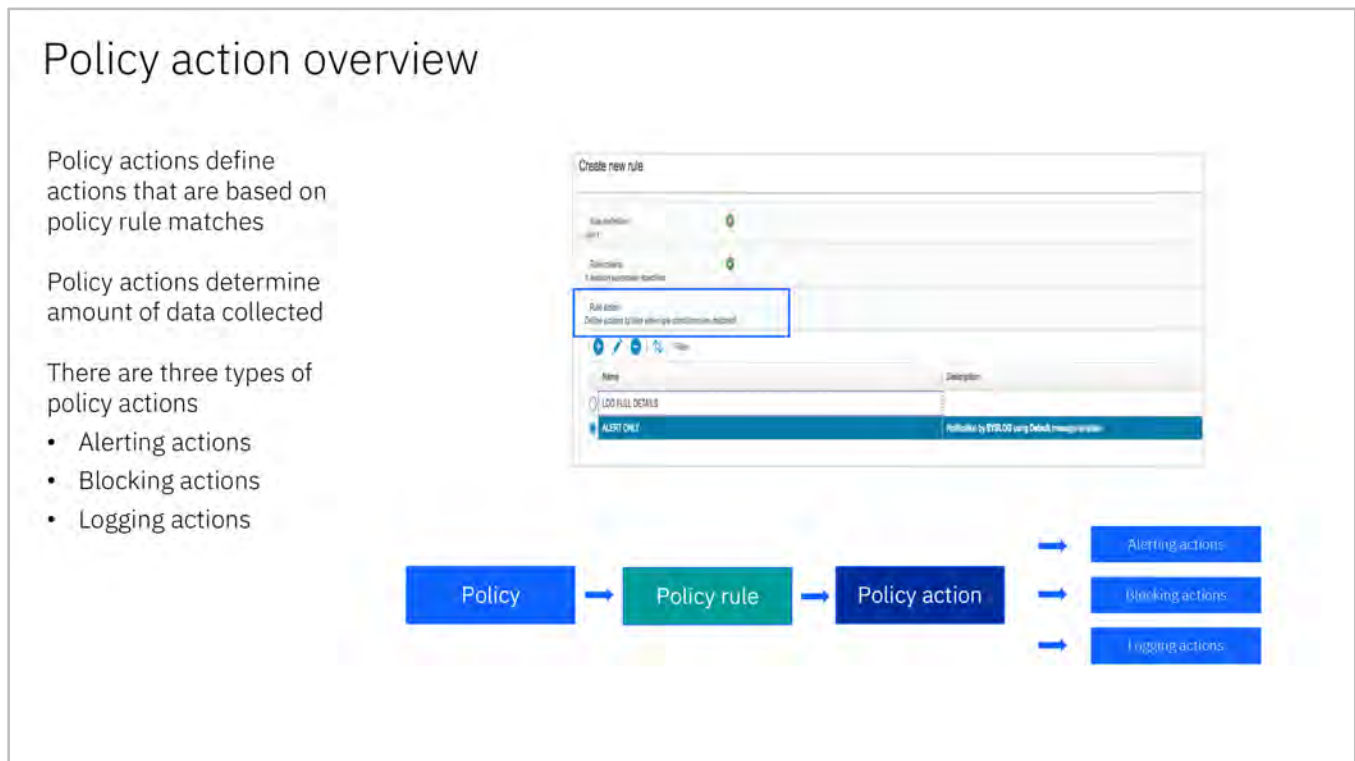
Policy actions

Figure: Policy actions



A Guardium rule can take several actions. This lesson defines the differences between the alerting, blocking, and logging policy action types.

Figure: Policy action overview

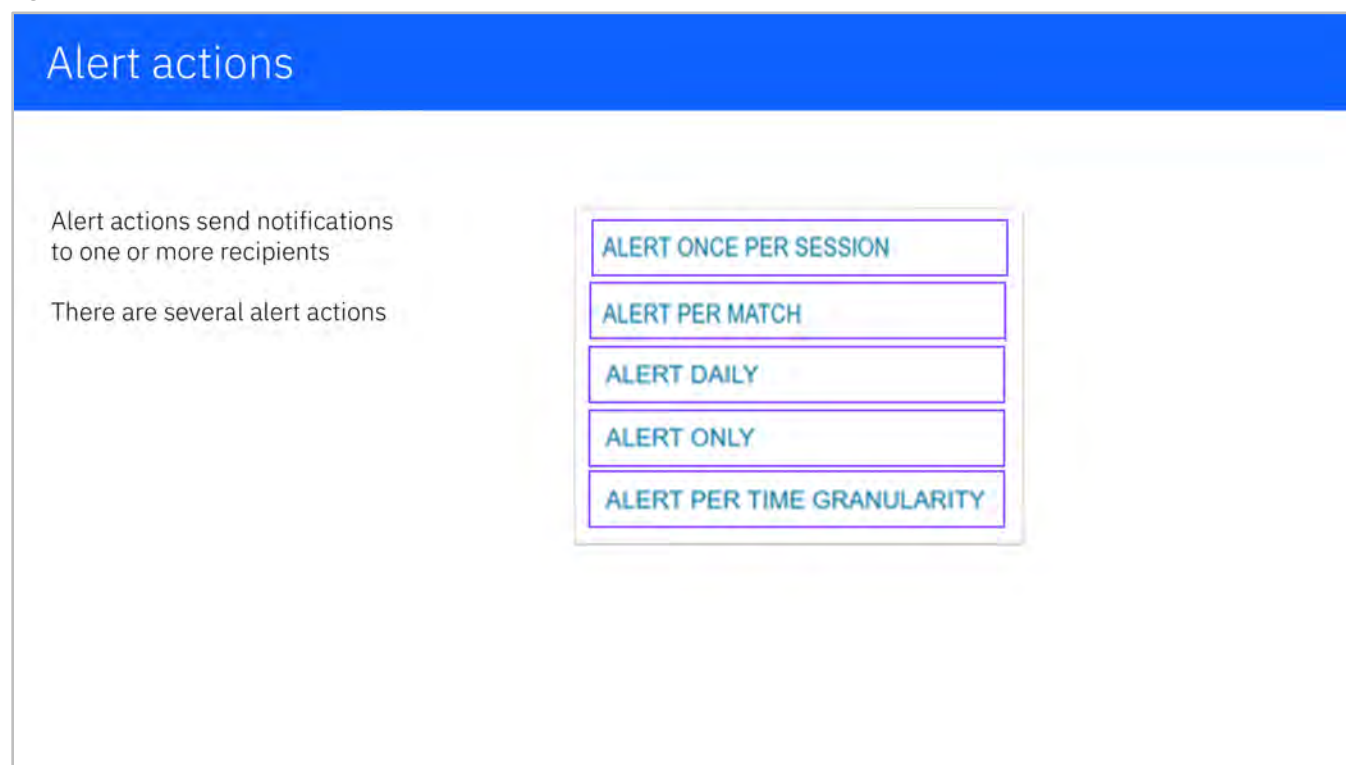


Policy actions are a subset of Guardium policies. Policy actions define what traffic is ignored, what traffic is logged, or which activities trigger an alert or block access to the database when the rule is matched.

Policy actions influence how much data comes into the Guardium appliance. Each rule in a policy defines a conditional action.

The action that is triggered by the rule can be a notification action, such as an email to one or more recipients. It can be a blocking action that disconnects the client's session. Or the event might be logged as a policy violation. You can develop custom actions to perform any tasks necessary for conditions that are unique to a particular environment or application.

Figure: Alert actions



Alert actions send notifications to one or more recipients.

For each alert action, multiple notifications can be sent, and the notifications can be a combination of one or more of the following notification types:

- **Email messages:** Addressed to Guardium users and are sent through the SMTP server that is configured for Guardium
- **SNMP traps:** Sent to the trap community that is configured for the Guardium appliance
- **Syslog messages:** Written to syslog
- **Custom notifications:** User-written notification handlers that are implemented as Java classes

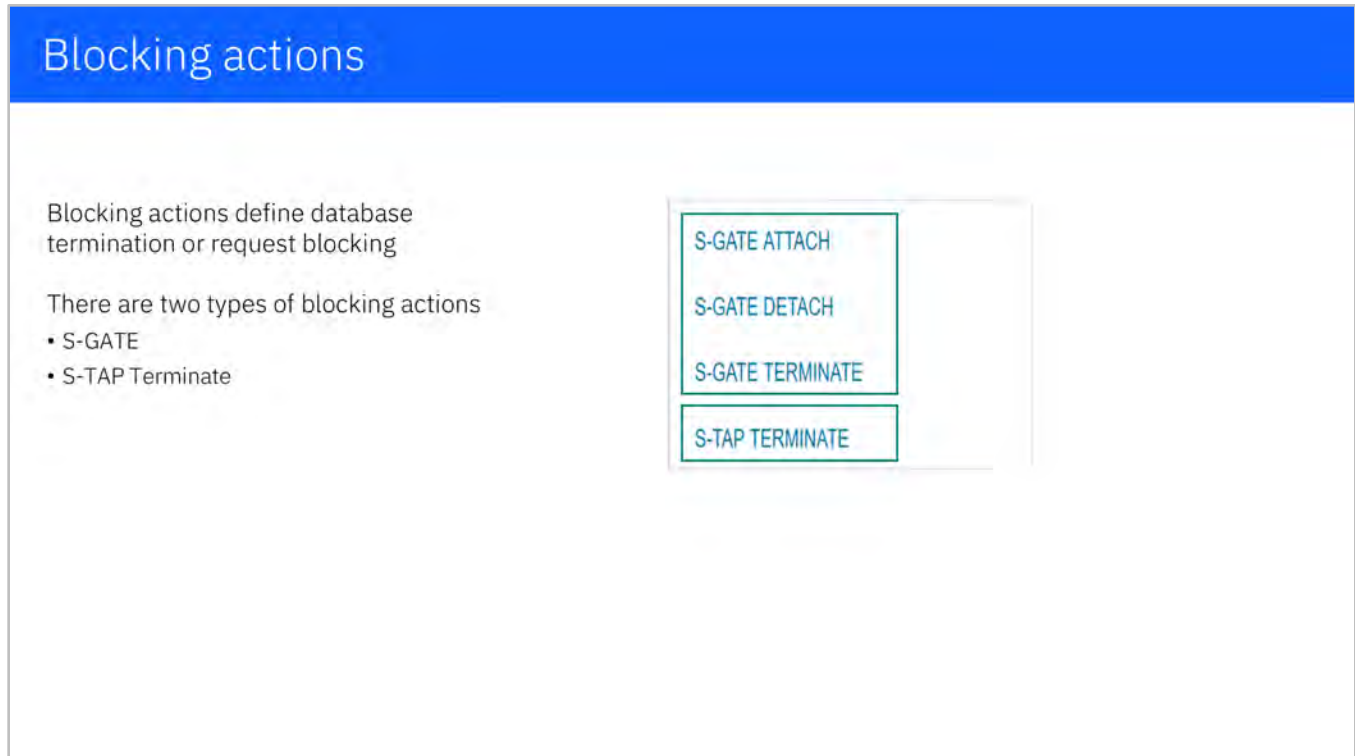
There are several types of alert actions with different granularities:

- **Alert Once Per Session:** This action sends notifications once for each session in which the rule is met. If you need to know that a certain event occurred, but you don't need to know about every instance of the event during a single session, use this action. For example, you might want a notification to be sent when a certain sensitive object is updated. But if a program updates thousands of instances of that object in a single session, you do not want thousands of notifications sent to the receivers of the alert.
- **Alert Per Match:** This action sends notifications each time the rule is satisfied. This action is appropriate for a condition that requires attention every time it occurs.
- **Alert Daily:** This action sends notifications only the first time that the rule is met each day. If you require alerts to be sent once each day, use this action.

- **Alert Only:** Messages go directly to /var/log/messages. If you require alerts to be sent to the Guardium System log only, use this action.
- **Alert Per Time Granularity:** This action sends notifications once per logging granularity period. If you require alerts to be sent once during the logging granularity period, use this action.

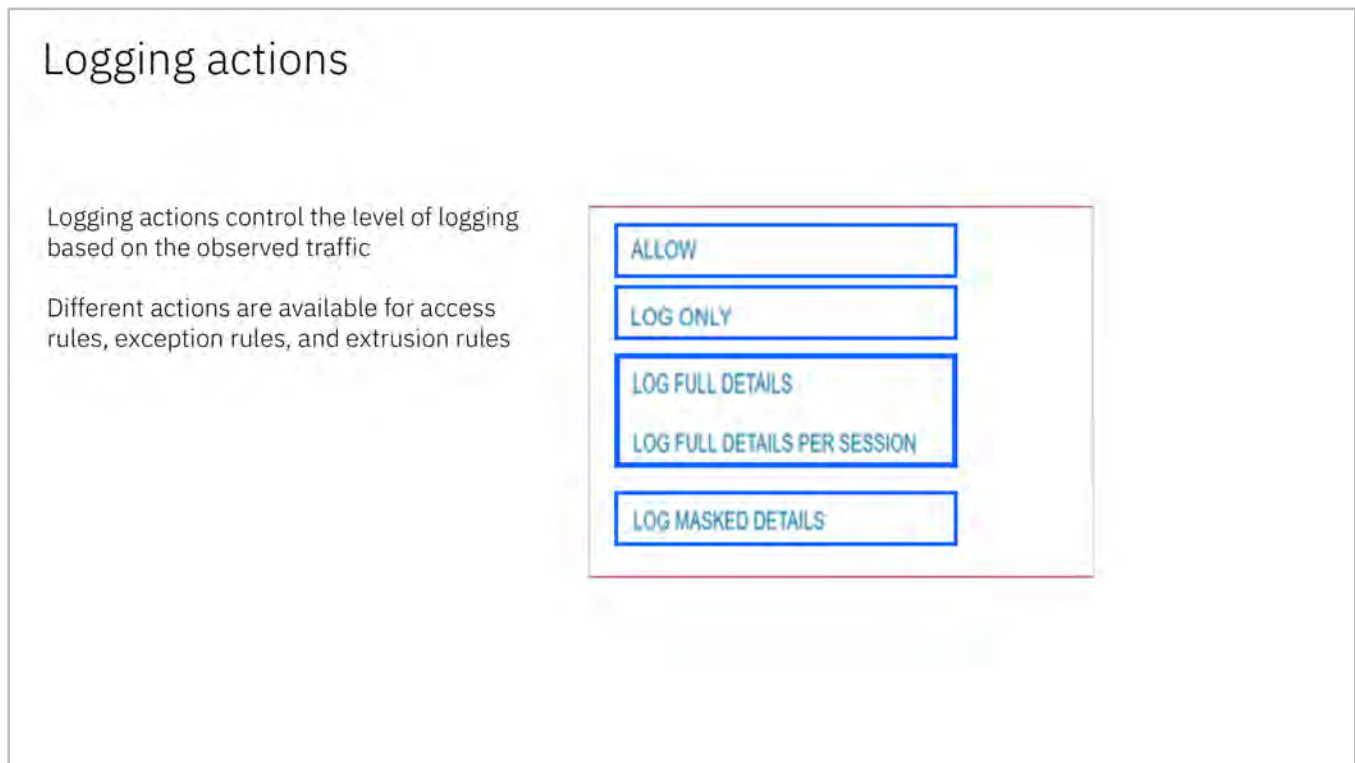
Note: Too many alerts can overwhelm the system, especially when the alert notification is sent to a SIEM.

Figure: Blocking actions



Blocking actions define database termination or request blocking. This rule action describes S-TAP and S-GATE Terminate rule actions. Blocking actions are useful when blocking or monitoring rogue requests.

Figure: Logging actions



Logging actions control the level of logging based on the observed traffic. You can combine logging actions.

Access rules, exception rules, and extrusion rules differ in which actions are available. For example, Log and Ignore actions are available for most policies, but the Audit Only action is available only for policies that use the Selective Audit Trail setting.

The common logging actions are:

- **Allow:** Log the matching construct, but do not log a policy violation.
- **Log Only:** Log the policy violation every time a rule is triggered
- **Log Full Details:** Log the full SQL string and the exact timestamp for the request. Consider whether there is a need for log full actions because they can quickly fill up the Guardium database.
- **Log Full Details per Session:** Log the full SQL string and exact timestamp for this request and for the remainder of the session.
- **Log Masked Details:** Log the full SQL for the request and replace values with question marks (?).

